

# Bonian Han

✉ bonianhan@u.boisestate.edu

## Education

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| <b>Hangzhou Dianzi University</b> <i>BS in Statistics</i>                        | <i>Sept 2021 - July 2025</i> |
| ◦ GPA: 3.48 (84.8/95)                                                            |                              |
| <b>New Jersey Institute of Technology</b> <i>PhD studies in Computer Science</i> | <i>2025 - 2026</i>           |
| ◦ GPA: 4.0; Advisor: <a href="#">Prof. Zhi Wei</a> ✉                             |                              |
| <b>Boise State University</b> <i>PhD in Computing (expected start)</i>           | <i>Fall 2026</i>             |
| ◦ Advisor: <a href="#">Prof. Yu Zhang</a> ✉                                      |                              |

## Skills & Interests

**Research Interests:** Computer Vision, Continual Learning, Embodied AI, Trustworthy Machine Learning (Model Calibration and Uncertainty Quantification), Neural Representations for Spatial Transcriptomics

**Skills:** Python, C/C++, PyTorch, SQL, R, Linux, Bash, Git, Docker

**Languages:** TOEFL: 91, GRE: 326 (Verbal: 156, 69th percentile; Quantitative: 170, 92nd)

## Publications

1. J. Zhang, **B. Han**, G. Liang, and Y. Zhang. “Rethinking Learned Occupancy in Autonomous Active Mapping with Observation-Gated Filtering.” *IEEE/RSJ IROS 2026 Workshop on Space Exploration and Sustained Operations Beyond Earth*, 2026. **Oral; Best Paper Runner-Up Award.**
2. **B. Han**, C. Qi, P. Musialski, and Z. Wei. “INST-Align: Implicit Neural Alignment for Spatial Transcriptomics via Canonical Expression Fields.” *Medical Image Computing and Computer Assisted Intervention (MICCAI)*, 2026. Accepted. [Paper](#)✉.
3. **B. Han**, Y. P. Masupalli, X. Xing, and G. Liang. “Improving Medical Imaging Model Calibration through Probabilistic Embedding.” *IEEE International Conference on Big Data*, 2024. DOI: [10.1109/BigData62323.2024.10825661](#)✉.
4. **B. Han**, C. Moran, J. Yang, Y. Lee, Z. Cao, and G. Liang. “Multi-Scale Self-Supervised Consistency Training for Trustworthy Medical Imaging Classification.” *46th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, 2024. Podium (acceptance rate approx. 28%). H5-Index: 48. DOI: [10.1109/EMBC53108.2024.10782322](#)✉.
5. J. Zulu, **B. Han**, I. Alsmadi, and G. Liang. “Contextualized Embedding-Based Approach for Enhanced SQL Injection Detection.” *62nd ACM SouthEast Annual Conference*, 2024. DOI: [10.1145/3603287.3651187](#)✉.

## Experience

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| <b>Research with Dr. Gongbo Liang, Texas A&amp;M University-San Antonio</b>                                                                                                                                         | <i>Nov 2023 - Nov 2024</i>  |
| ◦ Conducted computer vision research on medical image analysis, model calibration, and feature extraction to reduce model bias. Three papers accepted: <i>IEEE EMBC</i> , <i>ACMSE</i> , and <i>IEEE Big Data</i> . |                             |
| <b>Research with Prof. Zhi Wei, New Jersey Institute of Technology</b>                                                                                                                                              | <i>Sept 2025 - 2026</i>     |
| ◦ Studied spatial transcriptomics and neural representations. Worked on INST-Align, coupling spatial alignment with a canonical expression field; accepted to MICCAI 2026.                                          |                             |
| <b>Working Ballkids Coach &amp; Manager</b>                                                                                                                                                                         | <i>Sept 2024</i>            |
| ◦ Trained and managed ballkids (ages 9-14) at the Hangzhou Open ATP 250 and ATP Challenger; coordinated on-court schedules across departments and ensured smooth operations, court etiquette, and safety.           |                             |
| <b>Volunteer 19th Asian Games and Asian Para Games</b>                                                                                                                                                              | <i>Sept 2023 - Oct 2023</i> |
| ◦ Served as a Tennis Ballkids & Ceremony Assistant, managing on-court operations and assisting players and umpires; featured by CCTV and honored as an Outstanding and Excellent Volunteer.                         |                             |

## Awards & Honors

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| <b>Best Paper Runner-Up Award</b> , IEEE/RSJ IROS 2026 Workshop                           | 2026     |
| <b>Outstanding &amp; Excellent Volunteer</b> , National                                   | 2023     |
| <b>Second Prize</b> , Statistics Research Competition, Provincial (State)                 | 2023     |
| <b>Third Prize</b> , National College Students Mathematics Competition (Online Challenge) | 2022     |
| <b>Fifth Place</b> , University Basketball Annual Competition                             | May 2022 |